

Milestone 2 – Feature Engineering & Baseline Modeling

1. Objective

The objective of this milestone is to establish a high-performing baseline scanner source identification system using handcrafted feature engineering and classical machine learning models. This milestone focuses on extracting discriminative scanner-specific features and evaluating their effectiveness using baseline classifiers.

2. Feature Engineering

Feature extraction is performed at the image level, where each document image is represented by a single feature vector.

2.1 Frequency-Domain Features (FFT):

Frequency-domain analysis is carried out by computing radial Fast Fourier Transform (FFT) band energies from the images. The Fourier spectrum is divided into multiple radial frequency bands, and the average energy within each band is calculated to identify mechanical scanner noise.

2.2 Texture Features (Local Binary Patterns - LBP):

Local Binary Pattern (LBP) histograms are used to model fine-grained texture variations caused by different scanners. Uniform LBP encoding is applied, and normalized histograms are included as features to capture sensor-specific signatures

2.3 Feature Vector Construction:

All extracted features—including FFT band energies and LBP texture features—are concatenated to form a single feature vector for each image. These feature vectors are stored in CSV format and used as input for baseline model training.

3. Baseline Modeling

Two classical supervised machine learning models are used to evaluate baseline performance.

3.1 Random Forest Classifier

An ensemble-based model composed of multiple decision trees, robust to feature correlations and well-suited for heterogeneous handcrafted features.

3.2 Support Vector Machine (SVM):

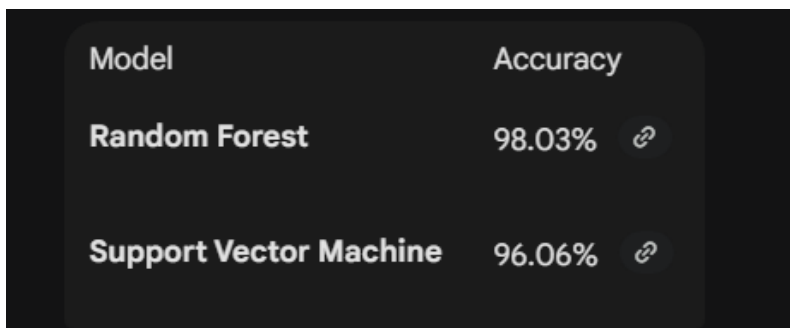
Implemented with a Radial Basis Function (RBF) kernel, capable of learning non-linear decision boundaries. A StandardScaler is applied to normalize feature distributions prior to training

Experimental Setup:

The dataset is split into 80% training and 20% testing.

5. Results

The baseline models demonstrate strong classification performance across the scanner classes.

A screenshot of a table with a dark background and light text. The table has two columns: 'Model' and 'Accuracy'. There are two rows of data. The first row shows 'Random Forest' with an accuracy of '98.03%' and a small circular icon with a right-pointing arrow. The second row shows 'Support Vector Machine' with an accuracy of '96.06%' and the same icon.

Model	Accuracy
Random Forest	98.03% 
Support Vector Machine	96.06% 

Random Forest Performance:

Achieved consistently high precision and recall across all scanner classes. HP and Epson V550 scanners demonstrated near-perfect classification.

SVM Performance:

Showed good generalization capability with weighted F1-scores of 0.96. Slightly higher confusion was observed among similar Canon scanner models compared to the Random Forest.

6. Testing

The system was validated using individual test samples to verify real-world performance.

Test Case Sample:

True Label: Canon220

RF Prediction: Canon220

SVM Prediction: Canon220

6. Conclusion

This milestone successfully establishes a high-accuracy baseline scanner identification system. The accuracies of 98.03% and 96.06% validate the handcrafted features and provide a strong foundation for further research and enhancement in subsequent milestones.
